AI-Powered EHR Assistant — Safety-Gated Agent

Agentic AI · Adewale Adeyinka Falade · LangGraph, LangChain, OpenAI (GPT-4o), SQLite

What It Does

A patient-facing assistant that explains a person's own health records — labs, medications, visit summaries — in plain language, while refusing anything clinically unsafe. It retrieves the relevant records, explains them at the patient's literacy level, and cites trusted health sources.

Architecture — LangGraph state machine (4 nodes)

1. **Policy** — every request first routed through a safety policy: answer, refuse, or escalate.
2. **Agent** — GPT-4o with bound tools decides which tools to call.
3. **Tool** — executes the selected tool(s) against the data layer.
4. **Final** — composes the plain-language, cited response.

Tools (11)

- **EHR retrieval:** patient profile, encounters, clinical notes, labs, medications, allergies.
- **Patient education:** plain-language lab explanations, medication education, trusted-source lookup.
- **Safety:** a policy_route tool that classifies each request.

Responsible-AI Design (the differentiator)

- **Refuses** unsafe requests — diagnosis, medication changes, dose calculations.
- **Escalates** red flags — chest pain, stroke signs, severe breathlessness, suicidal thoughts.
- **Answers** only education questions — explaining a lab, a visit, general health info.
- **Grounded & cited** — answers cite a curated allowlist (MedlinePlus/NIH, CDC, NHS, Mayo), not the open web.

Self-authored prototype. Patient database and reference tables were synthetic course data and are not included; the notebook shows the full agent design. Set your own OPENAI_API_KEY to run.